

MANA Operations Platform

Backend Implementation - Work Breakdown Structure

2 full-stack developers - 6 weeks build + 2 weeks migration/buffer - July 2026

Role	Owner	Focus
Dev A - Operations Spine	Developer 1	Inventory, allocation board, lock pipeline, pick slips, dashboards, notifications
Dev B - Data & Finance	Developer 2	Order intake, customers, vendor import, QBO, Gmail, credits, vendor recon, deploys
Shared	Both	Week-1 schema/type foundations, end-to-end QA (Week 6), data migration and cutover (Weeks 7-8)

Rule of thumb: Dev A owns everything a warehouse/ops person touches; Dev B owns everything that enters or leaves the system (orders in, files in, invoices/emails out). Merge checkpoints at the end of Weeks 2, 4 and 6 are demoable to Blake.

0. Already done - do not rebuild

The following is complete and verified (type-checks + lint clean). Tasks below build on top of this; nothing here should be re-implemented.

Area	What exists
UI - all 13 pages	Login, Allocation Board (multi-species contents, per-species fulfillment, split/merge, drag-and-drop), Order Intake (multi-species lines, SKU pricing + overrides), Inventory (mixed boxes, working CSV import parser + manual Add Lot), Pick Slips (species-grouped, PARTIAL flags), Customers, Finance Queue, Credits (counter modal with amount), Vendor Recon (Draft/Sent/Countered/Settled lifecycle + activity trail), 3 dashboards with GA-style date filter (wired: CEO + Finance KPIs scale by range), Notifications, Admin.
Database - Supabase	Migrations 0001-0005 applied (20 tables, enums, RLS, lock/unlock RPCs, seeds, admin user) PLUS schema v2 fixes file 20260702100001 (order_lines, box_contents, vendor_mappings + species codes, standing_orders, notification_prefs, purchase_orders, FK tightening, fulfillment + availability views, reworked lock_board). Run v2 once if not yet applied.
Auth	Real login (signInWithPassword), middleware guard on /mana/*, sign-out destroys session, first signup = admin, /api/admin/invite route (service role), guard against self role escalation.
Admin live data	Users/roles (read + assign role + invite), SKUs (read + Add SKU insert), Pricing tiers (read + edit terms/multiplier). Vendor Mappings and Integrations tabs still static.
Repo / tooling	Next.js 16 app in apps/web (inline-style convention), NestJS starter in apps/api (empty), supabase/migrations + reset.sql + README, ESLint + husky pre-commit, docx walkthrough in docs/.

Ground rules (both developers)

Definition of done: feature reads/writes Supabase (no local-state mocks left on the page), RLS verified for at least admin + operations + viewer, loading/empty/error states present, type-check + lint clean, demoed on the weekly checkpoint.

Conventions: keep the inline-style system (no Tailwind/CSS modules on mana pages); NUMERIC money/weights; all mutations through shared data hooks in apps/web/src/lib/; derive allocation - never store allocated totals; every finance/inventory mutation writes audit_log.

Safety (AGENTS.md): allocation, splits, QBO writes, migration parity are safety-critical - human review required; never remove audit/validation/idempotency; migrations must be reversible.

Branching: feature branches off feature/version2, PR review by the other dev within 24h, merge checkpoints at end of Weeks 2/4/6.

Blockers to raise immediately: QBO app approval, Google OAuth consent verification, access to real vendor packing-list samples and the live Google Sheets.

Week 1 - Foundations (pair Mon-Tue, then split)

Goal: v2 schema live, typed data layer, deploy pipeline, integration clocks started.

Both (Mon-Tue, pairing)

Task	Detail / files	Done when
Apply + verify schema v2	Run 20260702100001_schema_v2_fixes.sql on the project. Verify: order_lines_with_fulfillment returns Nobu 81/90; availability_by_species returns rows; lock_board('SFO') refuses in SQL editor (no auth) - correct behavior.	Both verification queries pass; screenshot in PR.
Generate DB types	supabase gen types typescript > apps/web/src/lib/database.types.ts; wire into both clients (client.ts / server.ts) as createClient.	No 'any' rows in new data code; tsc clean.
Shared data hooks	apps/web/src/lib/data/: useLots(), useOrders(), useCustomers(), useSkus(), useAvailability(), mutations assignContent(), splitContent(), mergeSplit(). Single supabase import point.	Hooks unit-smoke-tested against seeded data.
Decide apps/api role	Confirm: Next.js route handlers for user-facing writes; NestJS (apps/api) reserved for background jobs (QBO sync worker, Gmail poller) with service-role key + DATABASE_URL from .env.example. Document in supabase/README.	One-page ADR committed.

Dev A (Wed-Fri)

Task	Detail / files	Done when
Realtime plumbing	Subscription helper for box_contents / orders / order_lines channels; reconnect + cleanup on unmount; used by board + finance queue later.	Two browser windows see each other's seeded-row edits.
Profiles/session context	React context exposing current profile (name, role, initials) from profiles table; Nav user card shows real name instead of hard-coded 'Blake'; last_seen updated on load.	Nav shows logged-in user; viewer role hides admin nav item.

Dev B (Wed-Fri)

Task	Detail / files	Done when
Deploy pipeline	Vercel project for apps/web; env vars (SUPABASE_URL, ANON_KEY, SERVICE_ROLE_KEY); preview deploys per PR; production branch = master.	Login works on the deployed preview URL.
QBO application - START CLOCK	Create Intuit developer account + app (Accounting scope). Request production keys same day - approval takes 3-5 business days and Week 4 depends on it. Store sandbox creds in Vercel env.	Sandbox company connects via OAuth playground.
Google Cloud project	OAuth consent screen (internal), enable Gmail API, scopes gmail.send + gmail.readonly; store client id/secret; document redirect URIs for local + Vercel.	OAuth consent completes for a test account.

Week 2 - Operations spine + intake live

Dev A

Task	Detail / files	Done when
Inventory page on DB	Replace lotData seed state with useLots() (lots -> boxes -> box_contents). Status filter maps to lot_status incl. 'incoming'. Manual Add Lot drawer inserts lot + boxes + contents in one RPC or transaction-ish sequence; CSV import confirm writes the same instead of local state.	Imported CSV appears for the other dev after refresh; rows survive reload.
Allocation board on DB	Orders column from orders_with_fulfillment + order_lines_with_fulfillment; inventory column from lots/boxes/contents; click + drag assign writes box_contents.assigned_order_line_id (match species to the order line); optimistic update + rollback on error.	81/90 figure comes from the view, not client math.
Split / merge on DB	confirmSplit deletes original content row + inserts two rows sharing split_group (single RPC recommended: split_content(content_id, weight_a, line_a, line_b)); merge reverses it. Guard: cannot split/merge locked content.	Split survives reload; merge restores original weight exactly (NUMERIC).
Realtime board	Subscribe to box_contents changes; second user's assignments appear without refresh; presence chip ('2 online') from Supabase presence.	Two-browser demo at Friday checkpoint.

Dev B

Task	Detail / files	Done when
Customers page on DB	CRUD against customers + price_overrides + standing_orders tables; tier chip drives from customers.tier; overrides editor (add/remove per SKU).	New customer visible in order intake chips.
Order intake writes orders	Create order -> insert orders row + N order_lines; ship_date, carrier, location from customer defaults; ticket number = real order code (ORD-#### sequence). New-customer path inserts customers row (T2 default) first. Replace SKUS/PRICE_OVERRIDES consts with skus + price_overrides + pricing_tiers queries (single source: DB).	Order created on intake appears on Dev A's board with per-species lines.
Standing orders v1	List + toggle on customer drawer; 'Create from standing order' button on intake pre-fills lines from standing_orders.	One click recreates Nobu's Mon/Thu Ahi order.

CHECKPOINT W2: intake -> board -> assign -> split, all persisted, two users live.

Week 3 - Lock pipeline + vendor import server-side

Dev A

Task	Detail / files	Done when
Lock/unlock end-to-end	Lock button -> rpc('lock_board', location); banner state from board_locks (active); unlock confirm dialog -> rpc('unlock_board'). Handle errors (already locked, role denied). Board read-only while locked (contents locked flag).	Lock blocks edits for BOTH users instantly (realtime).
Pick slips from DB	Page lists pick_slips joined to orders/customers; render box_refs jsonb grouped by species (shape already produced by lock_board v2: box/species/grade/weight/lot/partial). Tabs become dynamic per generated slip; keep print CSS.	Slip shows exactly what was locked, incl. PARTIAL rows for splits/mixed boxes.
Slip PDF + auto-email	Server route renders slip to PDF (@react-pdf/renderer or puppeteer-core on Vercel); store in Supabase Storage (pick-slips bucket); on lock, send to warehouse address via Gmail API (gmail.send, token from integration_tokens via service role). Retry once; failures create a notifications row.	Locking SFO emails a PDF; pdf_url saved on pick_slips.
Availability strip	Board header strip from availability_by_species (available / allocated / incoming per species, filtered by location).	Numbers match a hand SQL check.

Dev B

Task	Detail / files	Done when
Vendor mappings admin -> DB	Replace static Mappings tab with vendor_mappings + vendor_species_codes CRUD (columns, uom lb/kg, species-code dictionary editor). Admin-only (RLS already gates writes).	Editing a mapping changes the next import's parsing.
Server-side import commit	Port the client CSV parser to a route handler: upload file (multipart) -> parse with mapping from DB -> return preview JSON; confirm endpoint inserts lot + boxes + contents atomically (plpgsql function import_lot(jsonb) recommended) + audit_log row; store original file in Storage (packing-lists bucket). Add .xlsx support via sheetjs.	Same UX as current UI but rows land in DB; original file downloadable.
Incoming lots / pre-sell	Lot can be created with status 'incoming' (no received_at); purchase_orders table minimal UI: create PO (vendor, species, expected lb/date); receiving an import can link po -> lot and flip status to received.	Board shows incoming inventory sellable against, per meeting requirement.

CHECKPOINT W3 (internal): import file -> allocate -> lock -> emailed PDF slip.

Week 4 - Finance queue + QBO

Dev A

Task	Detail / files	Done when
Notifications live	notifications table CRUD + realtime bell badge in Nav; prefs page writes notification_prefs. Emit events: shortage detected, lock done, import confirmed, invoice failed.	Bell updates without refresh; mark-read persists.
Shortage detection	On assignment change, compare order_lines fulfilled vs target for orders shipping <= tomorrow; insert 'shortage' notification once per line (dedupe).	Ops dashboard shortage card = live query, not seed.
Ops dashboard live	KPIS from queries: boxes available (contents unassigned), open orders, shortages, pick slips pending; ALLOC_BY_SPECIES from availability view; RECEIVING from lots received in the date-filter range (wire DateRangeFilter selection into the query).	Numbers change when Dev B creates an order.

Dev B

Task	Detail / files	Done when
Invoice generation on lock	DB trigger or post-lock server step: one invoices row per locked order (amount = sum of line weight x unit_price; unit_price captured on order_lines at intake). Status 'pending'.	Locking creates pending invoices visible in Finance Queue.
Finance Queue on DB	Page reads invoices joined customers/orders; counters (pending/failed/synced) live via realtime; approve button -> POST /api/qbo/invoice.	Counters move as invoices change state.
QBO push (sandbox)	OAuth connect flow storing tokens in integration_tokens (service role only); create invoice via QBO API with Request-Id = invoice id (idempotent); map customer -> qbo_customer_id (create if missing), sku -> qbo_item; on success store qbo_ref + synced_at, on failure store err_msg + status failed + notification. Refresh-token rotation handled.	Approve twice = one QBO invoice (idempotency proven); failure path renders on card.
Retry worker (apps/api)	NestJS cron (every 5 min): retry failed syncs with backoff, max 5 attempts; structured logs.	Kill network mid-sync -> invoice recovers to synced automatically.

CHECKPOINT W4: full loop - import -> allocate -> lock -> slip emailed -> invoice in QBO sandbox.

Week 5 - Credits, downgrades, vendor reconciliation

Dev A

Task	Detail / files	Done when
Credits page on DB	credit_claims CRUD; new-claim drawer selects real order/lot/boxes (FKs lot_id, vendor_id auto-derived from the lot - the meeting's 'system identifies vendor/box/lot/fish'); counter modal writes counter_amount; approve -> status approved + audit_log.	Claim created against a real lot shows vendor auto-linked.
Downgrades on DB	downgrades CRUD (box_id/box_content_id FKs); applying a downgrade adjusts the content's grade and logs impact; dispute status flow.	Downgrade appears as deduction candidate in vendor recon.
Approve -> QBO credit memo	Approve claim calls Dev B's /api/qbo/credit-memo (linked to original invoice); qbo_ref stored; failure = notification + failed badge.	Credit memo visible in QBO sandbox against the right invoice.
Finance + CEO dashboards live	Finance KPIs (invoiced in range, AR from invoices, failed syncs, open credits) + AR aging from due dates; CEO revenue/boxes/margin from real aggregates; date filter drives queries incl. comparison period (two queries, delta rendered).	Custom range + compare shows real deltas.

Dev B

Task	Detail / files	Done when
Statement builder	Generate vendor_statements + statement_lines per vendor/lot at closeout: sold lines from locked contents (customer, weight, cost-basis rate), less downgrade credits + approved claims linked to that lot; net due computed in SQL.	VR numbers reproduce the current Kona mock from real data.
Recon lifecycle on DB	Statuses draft/sent/countered/settled persist on vendor_statements (sent_at/settled_at); activity trail from audit_log entries; settle locks the statement (no edits).	Lifecycle survives reload; settled statement immutable.
Email statement + capture reply	Send statement PDF via Gmail; store gmail thread id; NestJS poller (gmail.readonly) watches the thread + vendor address, saves replies to vendor_emails linked to the statement (parse_status pending). Manual 'mark countered' remains the human decision.	Vendor reply appears on the statement page within one poll cycle.

CHECKPOINT W5 (internal): claim -> approve -> credit memo -> statement deduction -> settled.

Week 6 - Hardening + full-loop QA (both)

Task	Detail / files	Done when
E2E scenario matrix (Both)	Scripted run: multi-species order incl. new customer; mixed-box + split allocation; shortage flag; lock; unlock + relock; slip email; invoice sync + forced failure + retry; claim counter then approve; downgrade; statement settle. Repeat concurrently as two users.	Matrix doc with pass/fail; all pass.
RLS role audit (Dev A)	Log in as all six roles; verify each page's read/write matches supabase/README matrix; attempt forbidden writes via console to prove policy blocks.	Denied-by-policy screenshots per role.
Error/empty/loading sweep (Dev A)	Every page: skeleton or spinner, empty-state copy, error toast with retry. Print CSS re-QA on slips + statements (Chrome/Edge).	No blank white panels anywhere.
QBO production swap (Dev B)	Swap sandbox -> production keys (approved in W1); reconnect OAuth as Blake's QBO; verify customer/item mapping against real chart; DO NOT push real invoices until cutover - feature flag QBO_LIVE=false.	Production OAuth connects; flag verified off.
Perf + indexes (Dev B)	EXPLAIN the board query with 1k contents; add missing indexes; paginate inventory (50 lots).	Board interactive < 500ms with 1k rows seeded.
Admin gaps (split)	A: customer-tier assign modal writes customers.tier. B: integrations tab shows real connection status (QBO token present, Gmail token present, storage ok).	No placeholder modals left except future integrations.

CHECKPOINT W6: MVP feature-complete. Weeks 7-8 are migration + buffer, not new features.

Weeks 7-8 - Data migration, pilot, cutover (both, safety-critical)

AGENTS.md: migration parity requires human sign-off and reversibility. Sheets remain source of truth until Blake approves cutover.

Task	Detail / files	Done when
Sheet inventory + mapping doc (Both, W7 Mon-Tue)	Export every live Google Sheet (orders, packing lists/inbound tabs, inventory tab, recon sheets); document column -> table.column mapping incl. species-code and lb/kg cases; flag unmappable rows for Blanca.	Mapping doc reviewed by Blake/Blanca.
Migration scripts (Dev B)	Node scripts: customers -> customers/overrides/standing_orders; historical lots/boxes -> lots/boxes/box_contents (status shipped); open items -> live rows; QBO customer ids matched by display name. Idempotent (upsert by natural keys) + rollback script per entity.	Dry-run on staging project passes twice (idempotent).
Parity checks (Dev A)	Automated report: row counts, total lb per lot, open-order totals, AR totals vs sheets/QBO; discrepancies listed with row refs. Human sign-off gate.	Parity report signed by Blake (business) + dev (technical).
Pilot week (Both, W8)	Team runs MANA in parallel with sheets for one real cycle; daily 15-min triage; fix fast; train Blanca/Sid (Loom videos per workflow).	One full real lot lifecycle completed in MANA.

Task	Detail / files	Done when
Cutover (Both, end W8)	Blake approves; QBO_LIVE=true; sheets frozen read-only; hypercare rotation for week 9.	First real invoice syncs to production QBO.
Stretch (only if green)	Gmail auto-ingest of vendor packing lists into import preview; SMS via Twilio for shortage alerts; CSV exports on tables; LAX location scoping in RLS.	Explicitly out of the 8-week promise.

Ownership map (quick reference)

Module / concern	Owner	Weeks
Schema v2 verify, DB types, data hooks, ADR	Both	1
Realtime helper, profile context	Dev A	1
Vercel deploys, QBO app, Google OAuth	Dev B	1
Inventory + import commit UI, board, split/merge, realtime board	Dev A	2
Customers, order intake writes, pricing from DB, standing orders	Dev B	2
Lock/unlock, pick slips, slip PDF + email, availability strip	Dev A	3
Mappings admin, server import + xlsx, incoming lots / POs	Dev B	3
Notifications, shortage engine, ops dashboard	Dev A	4
Invoice generation, finance queue, QBO push + retry worker	Dev B	4
Credits/downgrades, credit-memo hookup, finance + CEO dashboards	Dev A	5
Statement builder, recon lifecycle persistence, Gmail send/reply capture	Dev B	5
E2E matrix, RLS audit, states sweep / prod QBO, perf, admin gaps	Both (split)	6
Migration scripts + rollback	Dev B	7
Parity checks + sign-off report	Dev A	7
Pilot, training, cutover, hypercare	Both	8

Risk register

Risk	Impact	Mitigation
QBO app approval delayed past Week 3	Week 4 slips	Applied Day 1; sandbox unblocks all build work; swap weeks 4 and 5 if needed.
Google OAuth verification friction (readonly scope)	Reply-capture (W5) slips	gmail.send is enough for MVP emails; reply capture degrades to manual paste.
Real vendor files do not match seeded mappings	Import rework	Get 3 real sample files from Blake in Week 1; mappings are data, not code.
Sheet data quality (typos, merged cells, unit mixes)	Migration overruns	Weeks 7-8 reserved; unmappable rows go to Blanca queue, not guessed.
Concurrent allocation conflicts	Double-assigned fish	Single-row updates + realtime; optional: optimistic-lock check on assigned_order_line_id IS NULL in the UPDATE where-clause.
Scope creep from weekly demos	Timeline slips	New asks land in the stretch list; Phase 1 modifications already folded in.

Total: 6 build weeks + 2 migration/buffer weeks. Everything in Section 0 is already delivered; every task above lists its acceptance test so hand-off between developers is unambiguous.